

Unified Hybrid Telemetry System for Motion, Machine, and Environmental Data Capture Using Decentralized Node Architecture

Abstract:

A modular telemetry platform combining human motion capture, machine system data (via CAN bus), environmental sensing, and spatial tracking into a unified, decentralized network.

The system consists of ESP32-based wearable sensor nodes equipped with inertial measurement units (IMUs), environmental sensors, and local data storage. These nodes can also connect directly to machine systems through CAN bus or serial interfaces. Each node logs data locally and transmits structured packets to a central hub (e.g., Raspberry Pi), which then relays the data via wireless networks such as LoRaWAN, Wi-Fi, or LTE.

The system is capable of syncing with spatial simulation engines such as Unreal Engine to enable real-time or post-event simulation, digital twin rendering, or forensic reconstruction. Optional drone-mounted nodes collect aerial data to extend spatial context.

The platform is designed for tactical training, environmental mapping, industrial diagnostics, forensic analysis, and historical preservation. It is modular, scalable, and resilient in disconnected environments.

Claims:

1. A modular telemetry system comprising a plurality of sensor nodes, each equipped with a microcontroller, inertial measurement unit (IMU), optional environmental sensor, and communication interface, wherein each node is configured to collect data, store it locally, and transmit it to a central processing hub wirelessly or via serial connection.
2. The system of Claim 1, wherein at least one node is connected to a machine or vehicle via a Controller Area Network (CAN bus) or equivalent serial interface, and is configured to receive and log machine diagnostic or operational data in real time.
3. The system of Claim 1, wherein the central processing hub is configured to aggregate data from wearable human-mounted nodes, vehicle telemetry sources, environmental sensors, and aerial devices, and to transmit structured data to a remote server or simulation engine over LoRaWAN, Wi-Fi, or LTE.
4. The system of Claim 1, wherein data collected is synchronized with a spatial engine (e.g., Unreal Engine) to enable real-time or post-event digital twin reconstruction.
5. The system of Claim 1, wherein aerial devices (e.g., drones) are equipped with compatible sensor nodes and operate in a coordinated swarm to extend the spatial

awareness and environmental mapping capability of the system.

6. The system of Claim 1, wherein forensic data is collected, time-stamped, and stored to enable accurate post-incident reconstruction, motion analysis, or site diagnostics.
7. The system of Claim 1, wherein the telemetry data is used to facilitate historical restoration or preservation of physical spaces through movement re-enactment, digital twin modeling, and spatial tracking.
8. The system of Claim 1, wherein live data fusion supports tactical training simulations by integrating human, machine, and environment telemetry into a synchronized scenario, optionally with XR or VR interface compatibility.
9. The system of Claim 1, wherein any node can operate autonomously in disconnected environments, and later synchronize with a central system using time-stamped data logs and wireless re-integration protocols.
10. The system of Claim 1, wherein data captured is optionally tokenized, serialized, or encrypted for secure transmission, decentralized storage, or integration into a permissioned digital asset environment.

Functional Description:

The system is composed of wearable sensor nodes, vehicle-mounted CAN-enabled modules, environmental monitors, and optional drone-based devices. Data is logged locally and transmitted to a control hub for aggregation, processing, and network relay.

The control hub (e.g., Raspberry Pi) acts as a gateway between the sensors and spatial visualization engines. Data may be viewed live, stored for forensic analysis, or simulated in environments like Unreal Engine.

Use cases include tactical operations, disaster zone mapping, search and rescue, digital forensics, XR-based training, autonomous drone scouting, and digital preservation of motion through spatial memory.

Diagram Notes:

1. System Overview Diagram:

- Human figure with labeled wearable sensor nodes (head, wrists, ankles, spine)
- Vehicle icon with CAN-enabled data connection to a node
- Drone overhead labeled “Environmental Sensor Node”
- Arrows from all devices to a central hub (e.g., Raspberry Pi)
- Hub broadcasting data to “LoRa / Wi-Fi / LTE” icons
- Output arrows to a screen labeled “Simulation Engine (e.g., Unreal)”

2. **Data Flow Diagram:**

- Sensor Node → Control Hub → Network → Cloud / Digital Twin
- Vehicle Node → Hub → Spatial Visualization
- Drone Node → Hub → Real-Time Mapping
- Feedback loop shown from Simulation back to Hub (optional)

3. **Deployment Map (Optional):**

- Multiple users, vehicles, and drones shown across a field or environment
- Lines indicating mesh or LoRa network coverage
- Zones labeled for tactical ops, training, recon, or restoration